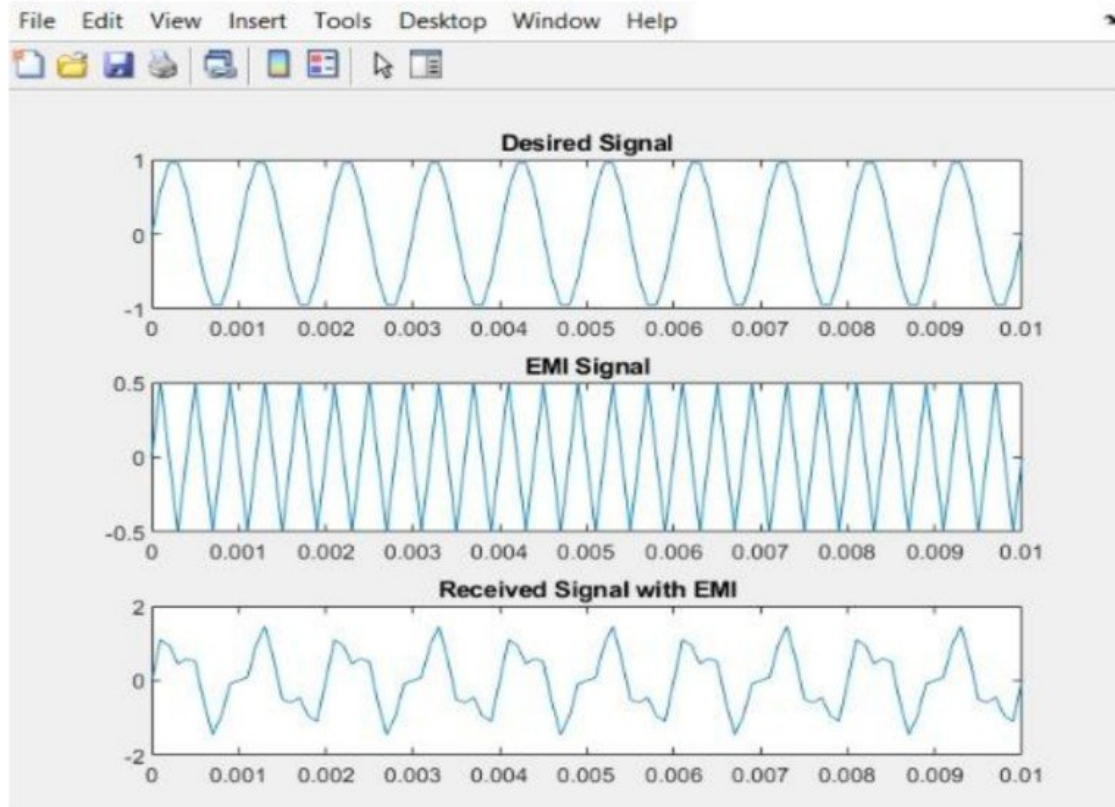
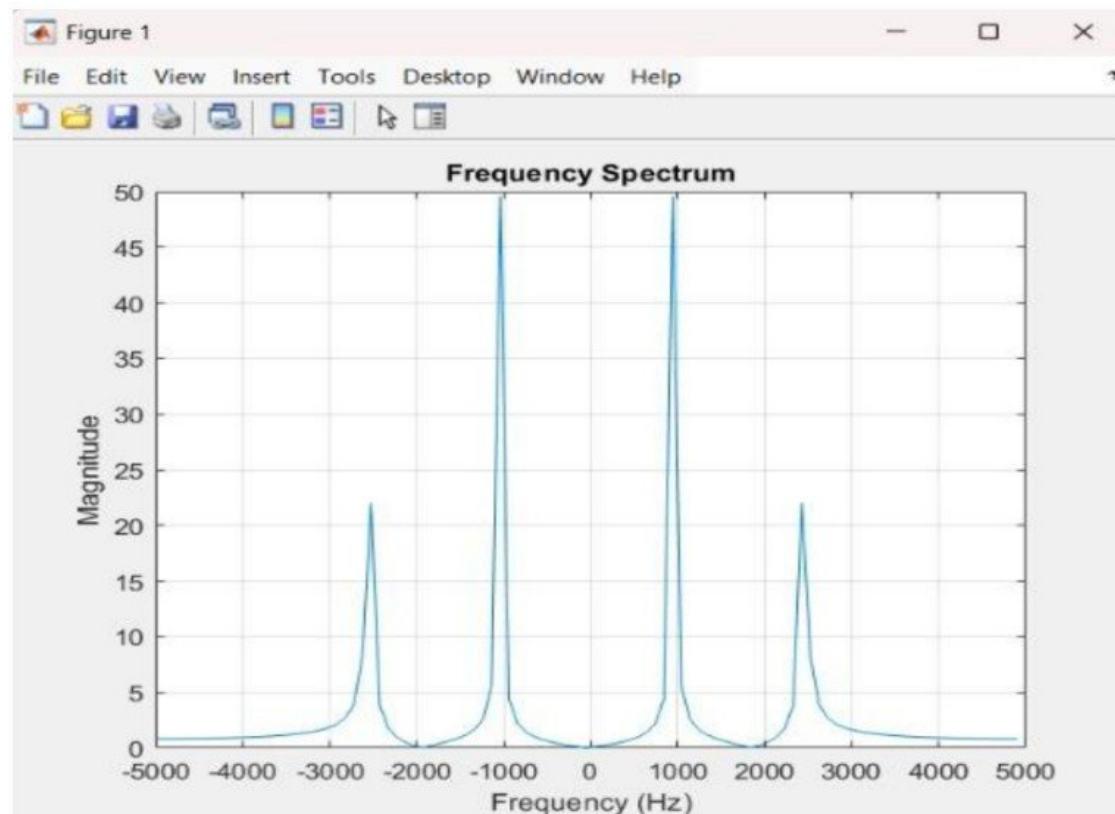


8a. Simulation of EMI Sources and Interference Analysis using MATLAB

Test case 1:



Test case 2:



Test case 3:

```
Command Window

>> untitled
>> untitled
SNR Before = 6.02 dB
SNR After = 33.29 dB
```

Test case 4:

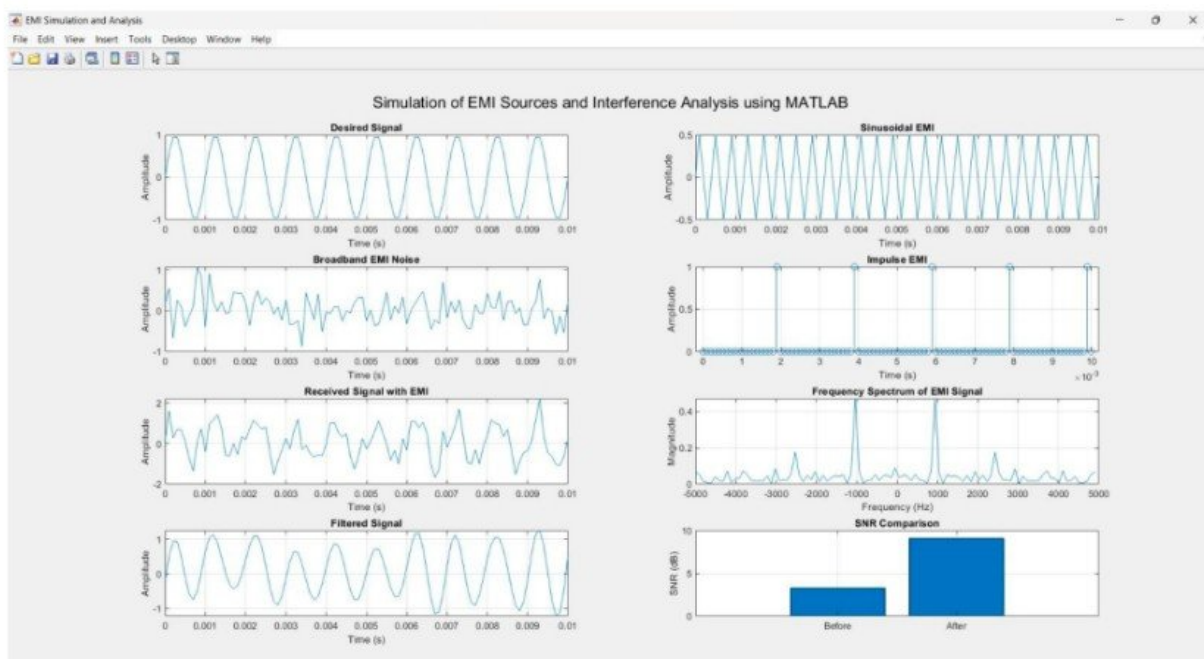
```
Command Window

-----
EMI ANALYSIS RESULTS
-----

Signal Frequency = 1000 Hz
EMI Frequency = 2500 Hz

Signal Power = 0.4950 W
Noise Power Before = 0.2328 W
Noise Power After = 0.0615 W

SNR Before Filtering = 3.28 dB
SNR After Filtering = 9.06 dB
SNR Improvement = 5.78 dB
```



8b. Shielding Effectiveness and EMC Compliance Analysis using MATLAB

Command Window

SHIELDING EFFECTIVENESS RESULTS

Maximum SE = 4419.37 dB

Minimum SE = 364.44 dB

System is EMC COMPLIANT

